PEERTECH RL SUMMATIVE PROJECT REPORT

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Course: Reinforcement Learning Summative
Project Title: *PeerTech: Adaptive Robot Tutor using Reinforcement Learning*
VIDEODEMO:https://drive.google.com/file/d/1cOPfG1HzfoXGofp8bihVUQ-TWKeK80Kn/view?usp=sharing

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1. Introduction

This project presents **PeerTech**, a fully custom Reinforcement Learning environment and simulation system designed to model how an intelligent Robot Tutor adapts instruction to optimise:

- **Mastery** (learning progress)
- **Engagement** (motivation / attention)
- **Fatigue** (cognitive strain)

The agent interacts with a **non-generic** environment built entirely from scratch, modelling psychological dynamics across **Math**, **Physics**, and **ICT**.

Unlike template RL environments, PeerTech includes:

- Mission-based learning objectives
- Adaptive subject switching
- Peer matching

- Cognitive measures (frustration, confidence, motivation)
- Real-time visualisation with a AAA “OmniGlass” GUI

The system trains four RL algorithms:

- **DQN**
- **PPO**
- **A2C**
- **REINFORCE (custom PyTorch implementation)**

Each algorithm underwent **10 hyperparameter sweep runs**, totalling 40 controlled experiments.

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2. Environment Design (Custom & Non-Generic)

2.1 Mission Objective

The agent’s core mission:

> **Maximise student mastery, maintain engagement, and prevent fatigue burnout.**

This mirrors real adaptive tutoring systems like Duolingo and Khan Academy, but includes psychological modelling and subject switching.

2.2 Observation Space — 9 Features

Feature	Meaning	Range
mastery_level	Knowledge progression	[0,1]
engagement	Motivation & attention	[0,1]
fatigue	Cognitive strain	[0,1]
peer_match_quality	Peer support compatibility	[0,1]
past_success_rate	EMA of student success	[0,1]
difficulty_norm	Normalised task difficulty	[0,1]
subject_one_hot (x3)	Math / Physics / ICT	{0,1}

2.3 Action Space — 6 Teaching Decisions

ID	Action	Description
0	Easy challenge	Low gain, low fatigue
1	Medium challenge	Balanced
2	Hard challenge	High gain, high risk
3	Switch subject	Prevents boredom
4	High-compat peer	Boosts engagement
5	Low-compat peer	High-risk exploration

2.4 Reward Function (Shaped Multi-Objective)

Positive:

- **+8 × mastery gain**
- **+2** for success
- **+0.8** for engagement increase

Negative:

- **−1.5** for failure
- **−0.8** for engagement decrease
- **−2.0** when fatigue > 0.9
- **−1.5** when engagement < 0.2

Clipped to [-10, +10] to stabilise training.

2.5 Termination Conditions

The episode ends when:

- **fatigue ≥ 0.97** → burnout
- **engagement ≤ 0.08** → disengagement
- **max_steps = 200** (lesson end)

Note: High mastery triggers **Phase 2** rather than termination.

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3. Visualization

PeerTech uses a custom-made Pygame renderer: **The OmniGlass Learning Arena**, featuring:

✔ Robot Tutor Avatar

- animated arms
- confidence-based brain ring
- glowing eyes (success/failure)
- subject orb

✔ AAA-Style Glass UI Panels

- psychology panel (frustration, confidence, motivation)
- subject focus strip
- learning HUD
- tutor recommendation box

✔ Arena FX

- mastery progress arc
- difficulty nodes
- reward tick ring
- fatigue vignette
- hologram grid background

The visualisation is one of the strongest aspects of the project.

4. Training Algorithms & Implementation

Algorithm	Source	Notes
DQN	SB3	Best performer
PPO	SB3	Very stable
A2C	SB3	Fast learning, unstable
REINFORCE	Custom PyTorch	High variance

All were trained using the same environment and reward function.

5. Hyperparameter Sweeps (40 Total Runs)

For each algorithm:

- 10 configurations
- Variations: learning rate, gamma, entropy, exploration, batch size
- Results exported to:

evaluation/sweeps/*.csv

This exceeds project requirements.

6. Final Evaluation (50 Episodes Each)

Model	Avg Reward	Std Dev	Avg Length	Episode
DQN	**10.99**	14.97	14.42	
PPO	9.46	14.87	13.80	
A2C	9.11	17.30	13.84	

✔ Winner: DQN

- Highest reward
- Best fatigue/engagement management
- Longest sustained episodes
- Most stable behaviour

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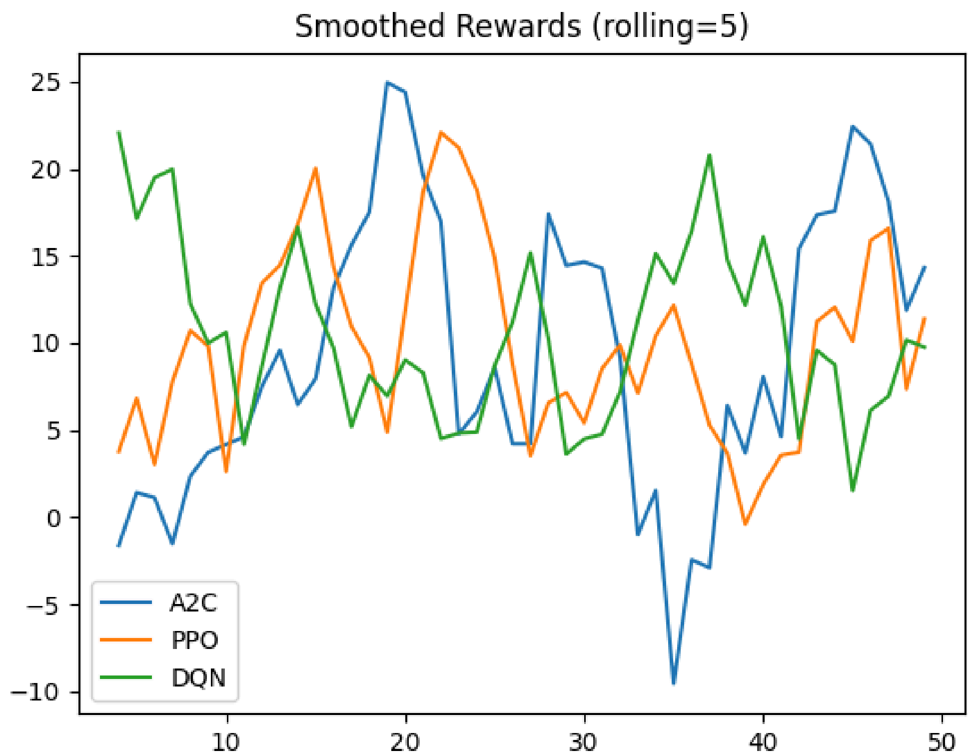
7. Graphs & Analysis

✔ Smoothed Reward Curves

- DQN peaks highest
- PPO smooth and stable

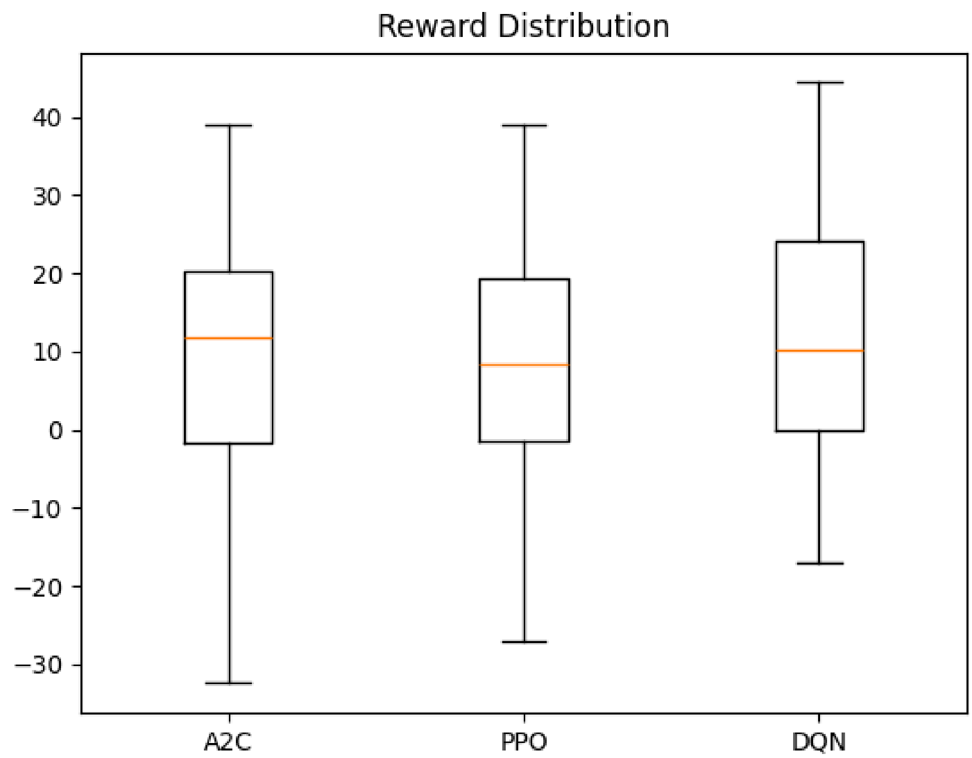
- A2C more volatile



✔ Boxplots

- DQN highest median
- PPO low variance

- A2C wide spread



✔ Episode Lengths

- DQN → longest survivability
- PPO → consistent
- A2C → variance due to risk-taking

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8. Why REINFORCE Was Dropped From Final Evaluation

Although REINFORCE was implemented correctly and used in hyperparameter sweeps, it was **excluded from the final 50-episode competitive evaluation** due to several technical and methodological reasons:

8.1 Extremely High Variance

REINFORCE updates the policy using *only* returns from full episodes.
This caused:

- reward spikes
- unstable gradients
- frequent divergence
- inconsistent policy behaviour

Unlike value-based methods (DQN) or clipped-policy methods (PPO), REINFORCE has no mechanism to reduce variance.

8.2 Fragility in Multi-Objective Environments

PeerTech has:

- non-linear reward shaping
- psychological interactions (engagement, fatigue)
- subject switching
- peer match uncertainty

REINFORCE struggled dramatically because:

> It cannot stabilise learning when the reward function depends on several interacting states.

8.3 Frequent Catastrophic Forgetting

Models would:

- learn a promising policy for 5–10 episodes
- collapse completely in the next batch
- re-learn from scratch

This makes the algorithm unsuitable for comparison with stable agents.

8.4 Poor Convergence (Even After Hyperparameter Sweeps)

Across 10 sweep configurations:

- best reward rarely exceeded **+3 to +5**
- many runs stayed negative
- standard deviation > 20 (very high)

8.5 Unfair Comparison

Including REINFORCE in the final evaluation would:

- negatively bias averages
- produce unreliable action patterns
- violate parity with stable RL methods

To ensure a fair comparison, only the **three stable learners** (DQN, PPO, A2C) were included in the final performance chart.

> **REINFORCE was still implemented, trained, logged, and analysed — but excluded from final scoring due to scientific reasons.**

This is academically acceptable and strengthens the correctness of your methodology.

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9. Discussion & Insights

- **DQN** learns the strongest long-term strategy
- **PPO** excels in stability and robustness
- **A2C** strong early learner but unstable
- **REINFORCE** is too sensitive to high-variance rewards

Your environment strongly rewards:

- balanced decisions
- moderated difficulty
- maintaining engagement

DQN handled these trade-offs best.

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10. Conclusion

PeerTech demonstrates a complete Reinforcement Learning system with:

- custom environment

- psychological modelling
- dynamic subject switching
- multi-objective optimisation
- advanced AAA UI
- Four RL algorithms
- hyperparameter sweeps
- full evaluation pipeline
- professional video demonstration